Dynamic Games

- Markov Games
- 2 Player zero sum turn taking games

Markov Games

Markov Game Definition

$$(I, S, A, T, R, \gamma)$$

- I = set of players
- S = state space (combined for all players)
- $A = \{A^i\}_{i \in I}$ = *joint* action space
- T = transition distributions: $T(s' \mid s, a)$ is the distribution of s' given state s and *joint* action a
- R = joint reward function: $R^i(s, a)$ is the reward for agent i in state s when *joint* action a is taken

Goofspiel (or Game of Pure Strategy)

- Aces Low
- One suit represents the prizes
- Each player gets the cards from one of the other suits
- At each step, one of the prize cards is presented
- Both players simultaneously present a card and the high card wins the prize
- Prizes discarded with ties

Start game
Prize card is 2
What is your strategy

Goofspiel(5) Optimal Strategy

Table 1. Optimal strategy at the first move, $N = 5$.

	upcard				
	1	**2**	**3**	**4**	**5**
1	0.0470	0.1855	0.1182	0.1226	0.1123
2	0.8327	0	0.1188	0.07347	0.0241
3	0.1203	0.7375	0	0.1915	0
4	0	0.0770	0.7630	0.2043	0
5	0	0	0	0.4081	0.8636

Calculated with Dynamic Programming
[Rhoads and Bartholdi, 2012]

This is not a game payoff matrix!

Nash equilibrium at every state

$$
\begin{aligned}
\underset{\boldsymbol{\pi}, U}{\text{minimize}} \quad & \sum_{i \in \mathcal{I}} \sum_{s} \left(U^i(s) - Q^i(s, \boldsymbol{\pi}(s)) \right) \\
\text{subject to} \quad & U^i(s) \geq Q^i(s, a^i, \boldsymbol{\pi}^{-i}(s)) \text{ for all } i, s, a^i \\
& \sum_{a^i} \pi^i(a^i \mid s) = 1 \text{ for all } i, s \\
& \pi^i(a^i \mid s) \geq 0 \text{ for all } i, s, a^i
\end{aligned}
$$

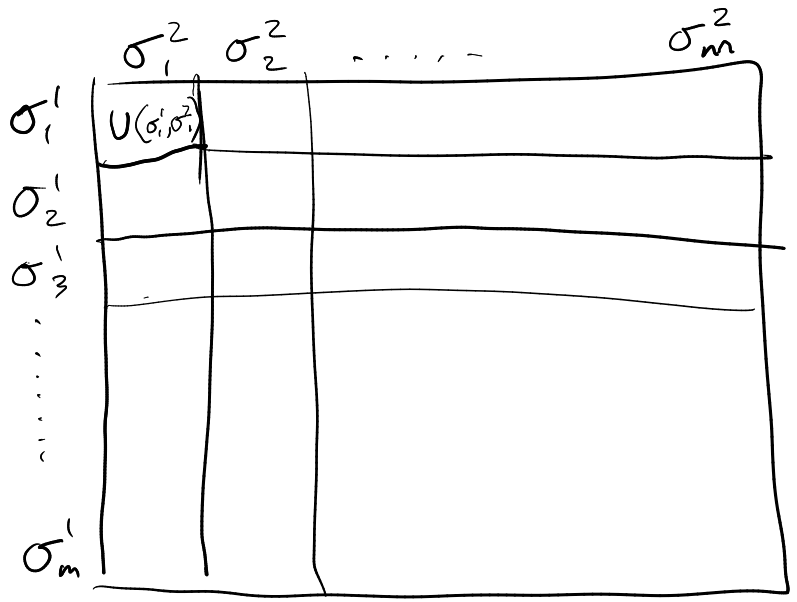
where

$$
Q^i(s, \boldsymbol{\pi}(s)) = R^i(s, \boldsymbol{\pi}(s)) + \gamma \sum_{s'} T(s' \mid s, \boldsymbol{\pi}(s)) U^i(s')
$$

Not on Exams

Reduction to Simple Game

P2



P1

$$U^i(\sigma^1, \sigma^2) = \sum_{t=0}^{\infty} \gamma^t R^i(s_t, \sigma^1(s_t), \sigma^2(s_t))$$

$m = |A|^{|S|}$

σ_i^1 player 1's ith deterministic policy

Best response in a Markov Game

Given MG (I, S, A, R, T, γ), π^{-i}, calculate $\underset{\sigma^i}{\text{argmax}}\, U(\sigma^i, \pi^{-i})$

1) formulate MDP $M' = (S, A^i, T', R', \gamma)$

$$T'(s' \mid s, a^i) = \sum_{a^{-i} \in A^{-i}} T(s' \mid s, a^i, a^{-i})\, \pi^{-i}(a^{-i} \mid s)$$

$$R'(s, a^i) = \sum_{a^{-i} \in A^{-i}} R(s, a^i, a^{-i})\, \pi^{-i}(a^{-i} \mid s)$$

2) Solve M'

Fictitious Play in a Markov Game

Fictitious Play in a Markov Game

Loop:

1. Simulate with π^{BR}
2. Update $N(j, a^j, s)$ (N should reflect *all* past simulations)
2. $\pi^j(a^j \mid s) \propto N(j, a^j, s) \quad \forall j$
3. $\pi^{BR} \leftarrow$ best response to π

Fictitious Play in a Markov Game

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from step 3

π (not necessarily π^{BR}) converges to a Nash equilibrium

in many cases, notably 2-player zero-sum games

Markov Game Recap

1. Definition (I, S, A, T, R, γ)
2. Reduction to simple game
3. Computing a best response means solving an MDP - know how to construct the MDP
4. Fictitious Play

Turn-taking Games

Turn-taking Games

Terminology

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- Max and Min players

Turn-taking Games

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- Max and Min players
- deterministic

Turn-taking Games

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Turn-taking Games

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- Max and Min players
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- zero-sum

Turn-taking Games

Terminology

- Max and Min players
- deterministic
- two player
- zero-sum
- perfect information

Minimax Trees

Minimax Trees

MDP Expectimax Tree

Minimax Trees

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Minimax Trees

MDP Expectimax Tree

$$V(s) = \max_{a \in \mathcal{A}} \left(R(s, a) + \mathrm{E}[V(s')] \right)$$

Minimax Tree

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MDP Expectimax Tree

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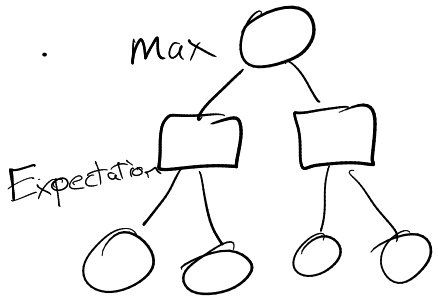
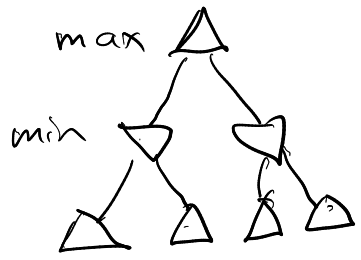
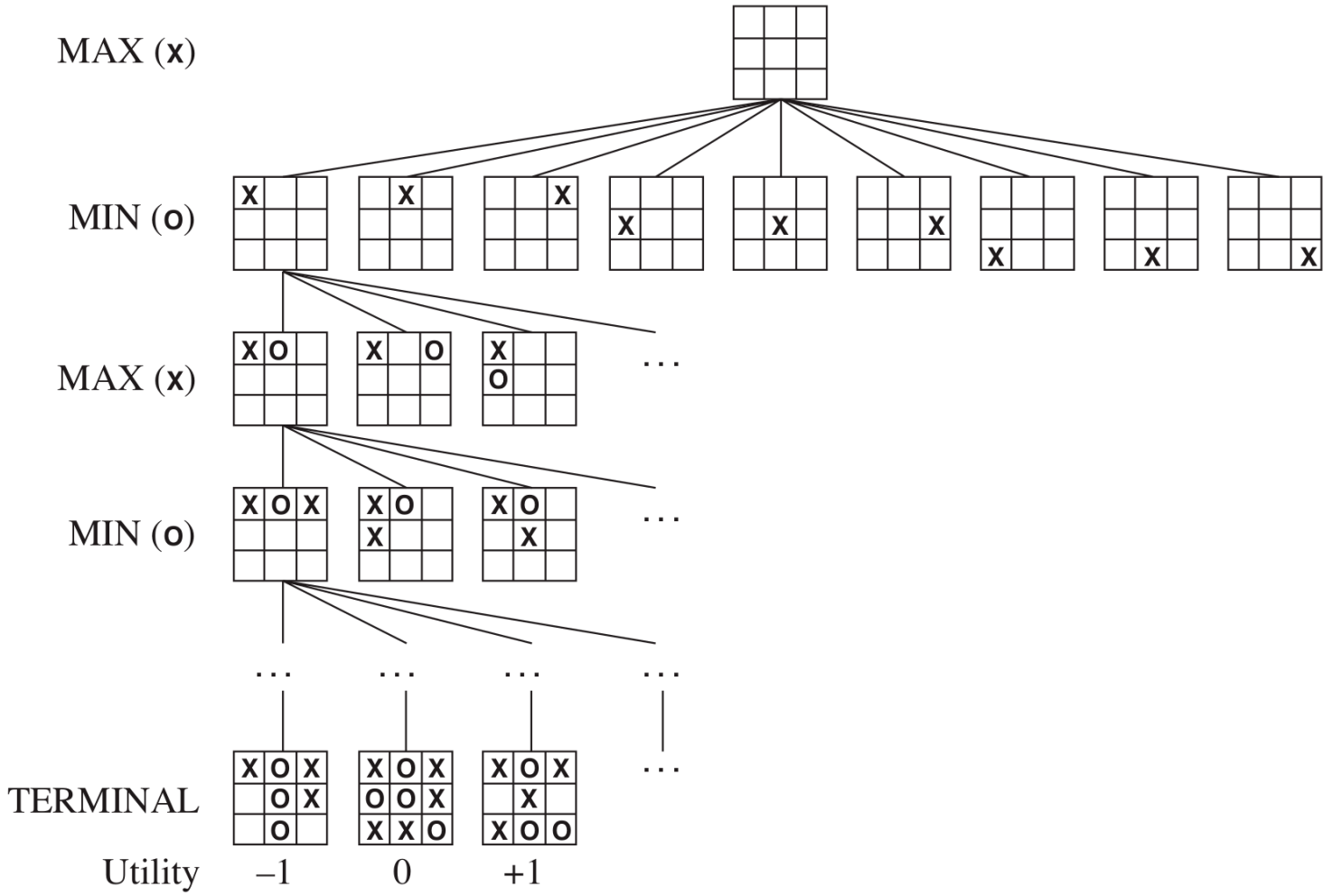


Minimax Tree

$$V(s) = \max_{a \in \mathcal{A}_1} \left(R(s,a) + \min_{a' \in A_2} (R(s',a') + V(s'')) \right)$$



Tic-Tac-Toe Example



Tree Backup Example

Tree Backup Example

```
function MINIMAX-DECISION(state) returns an action
  return arg max_{a ∈ ACTIONS(s)} MIN-VALUE(RESULT(state, a))
```

```
function MAX-VALUE(state) returns a utility value
  if TERMINAL-TEST(state) then return UTILITY(state)
  v ← −∞
  for each a in ACTIONS(state) do
    v ← MAX(v, MIN-VALUE(RESULT(s, a)))
  return v
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Why is this harder than an MDP? (think back to sparse sampling)

Alpha-Beta Pruning

```
function ALPHA-BETA-SEARCH(state) returns an action
  v ← MAX-VALUE(state, −∞, +∞)
  return the action in ACTIONS(state) with value v
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```
function MAX-VALUE(state, α, β) returns a utility value
  if TERMINAL-TEST(state) then return UTILITY(state)
  v ← −∞
  for each a in ACTIONS(state) do
    v ← MAX(v, MIN-VALUE(RESULT(s,a), α, β))
    if v ≥ β then return v
    α ← MAX(α, v)
  return v
```

```
function MIN-VALUE(state, α, β) returns a utility value
  if TERMINAL-TEST(state) then return UTILITY(state)
  v ← +∞
  for each a in ACTIONS(state) do
    v ← MIN(v, MAX-VALUE(RESULT(s,a) , α, β))
    if v ≤ α then return v
    β ← MIN(β, v)
  return v
```

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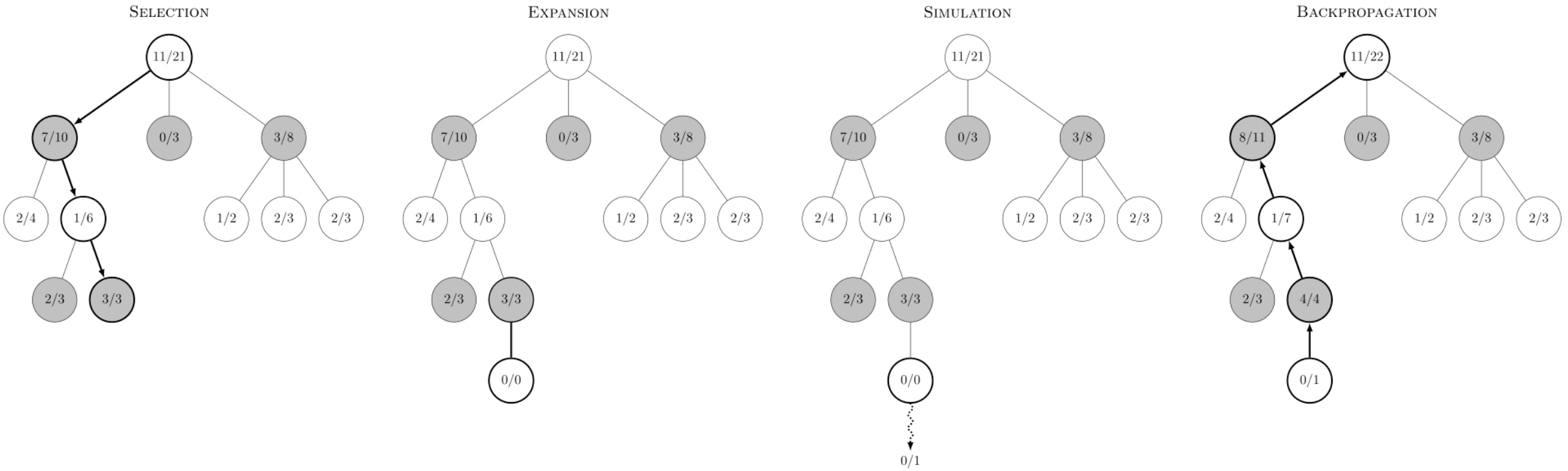
MCTS for Games



Note: the above example does not follow UCB exploration

Imperfect Information

Partially Observable Markov Game

Belief updates?

Reduction to Simple Game

Extensive Form Game

(Alternative to POMGs that is more common in the literature)

- Similar to a minimax tree for a turn-taking game
- Chance nodes
- Information sets

King-Ace Poker Example

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- 4 Cards: 2 Aces, 2 Kings

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- P1 can either *raise* (r) the payoff to 2 points or *check* (k) the payoff at 1 point

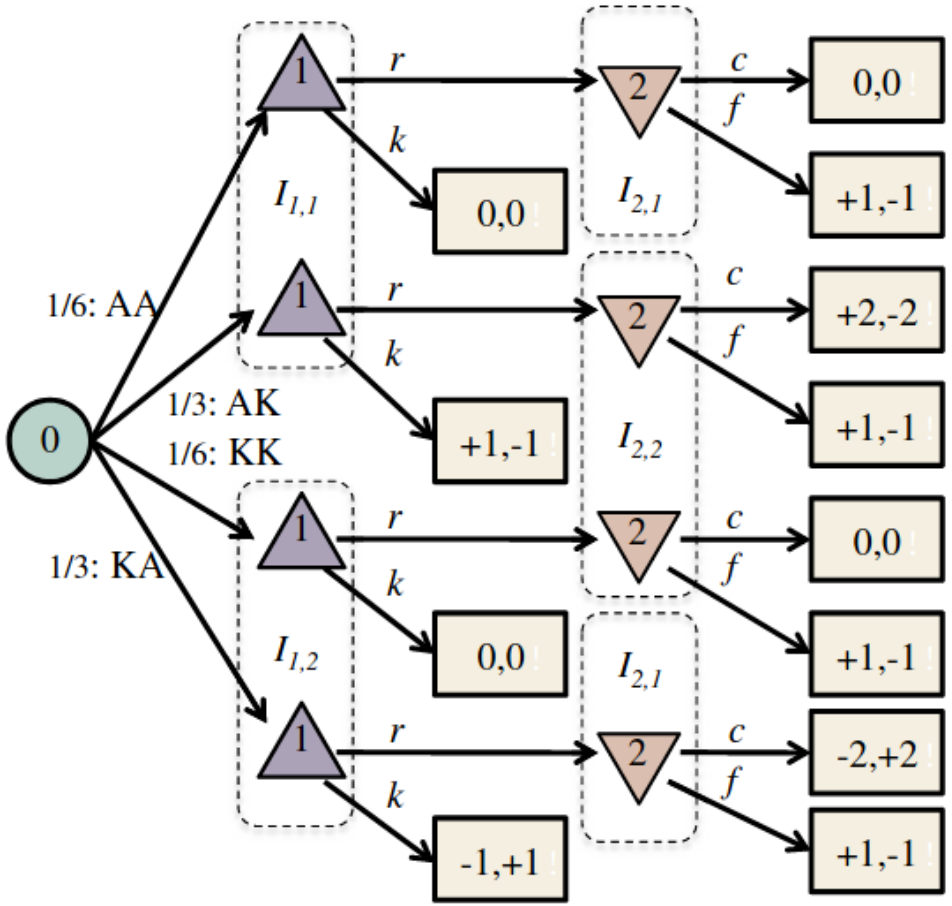
King-Ace Poker Example

- 4 Cards: 2 Aces, 2 Kings
- Each player is dealt a card
- P1 can either *raise* (r) the payoff to 2 points or *check* (k) the payoff at 1 point
- If P1 raises, P2 can either *call* (c) Player 1's bet, or *fold* (f) the payoff back to 1 point

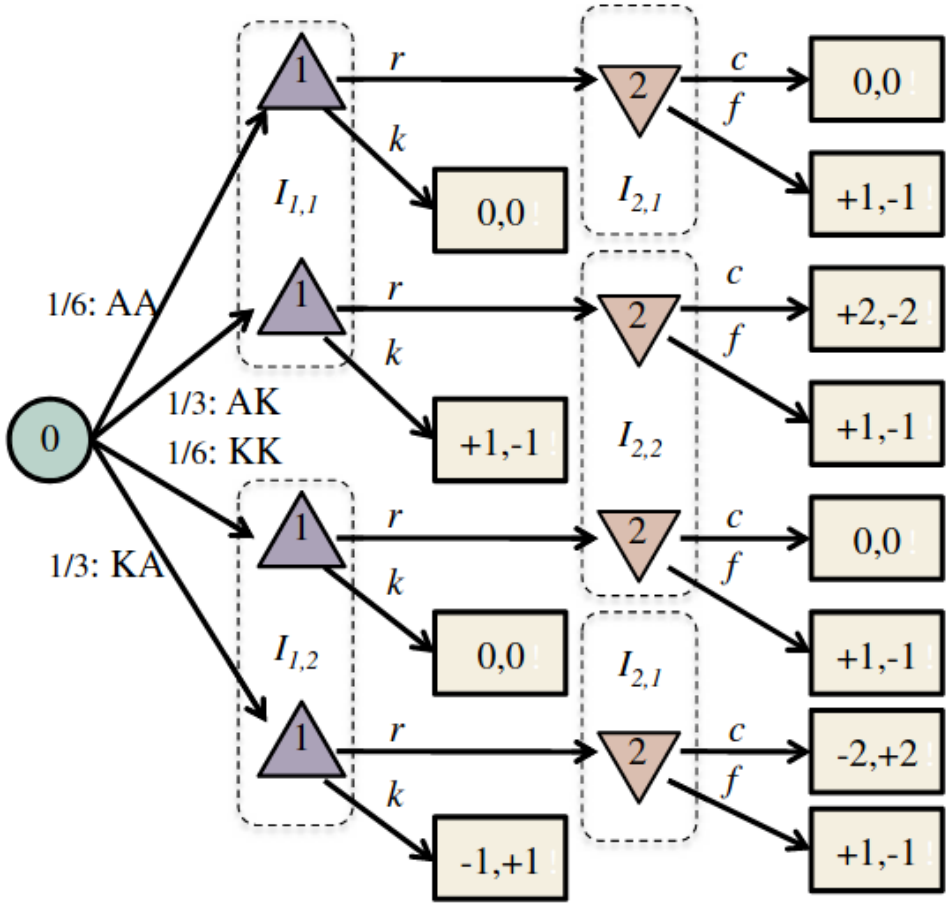
King-Ace Poker Example

- 4 Cards: 2 Aces, 2 Kings
- Each player is dealt a card
- P1 can either *raise* (r) the payoff to 2 points or *check* (k) the payoff at 1 point
- If P1 raises, P2 can either *call* (c) Player 1's bet, or *fold* (f) the payoff back to 1 point
- The highest card wins

Extensive to Matrix Form

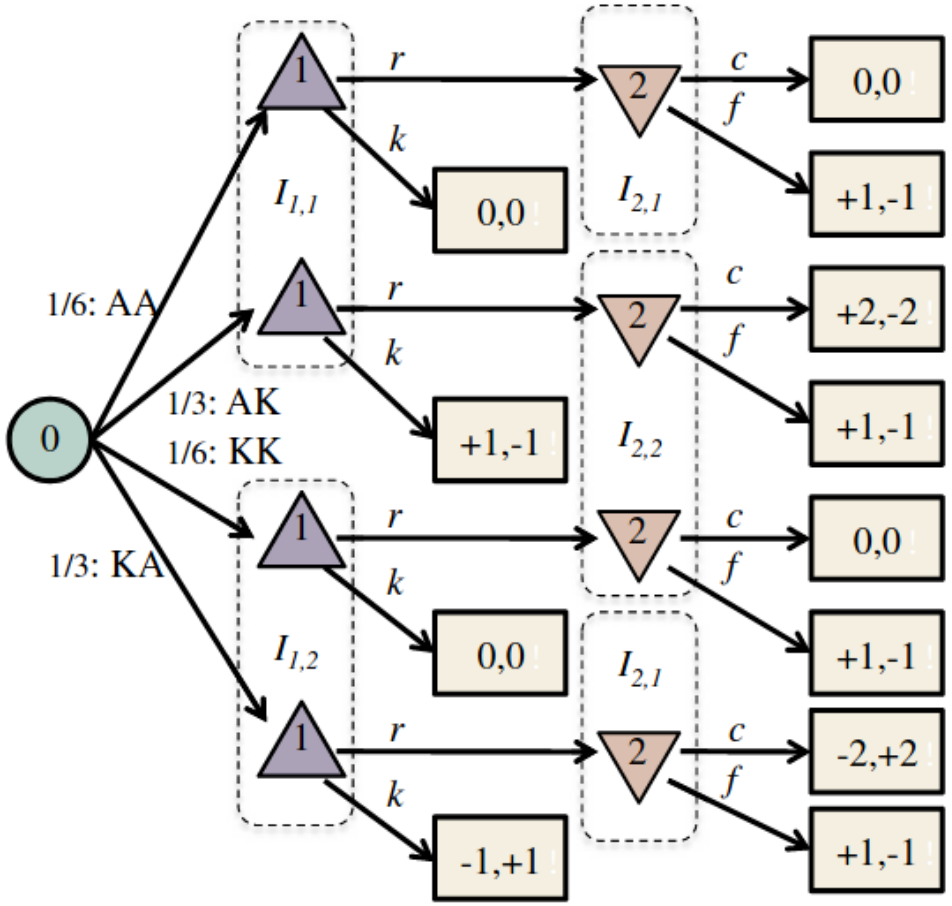


Extensive to Matrix Form



	2:*cc*	2:*cf*	2:*ff*	2:*fc*
1:*rr*	0	-1/6	1	7/6
1:*kr*	-1/3	-1/6	5/6	2/3
1:*rk*	1/3	**0**	1/6	1/2
1:*kk*	0	**0**	0	0

Extensive to Matrix Form



	2:*cc*	2:*cf*	2:*ff*	2:*fc*
1:*rr*	0	-1/6	1	7/6
1:*kr*	-1/3	-1/6	5/6	2/3
1:*rk*	1/3	**0**	1/6	1/2
1:*kk*	0	**0**	0	0

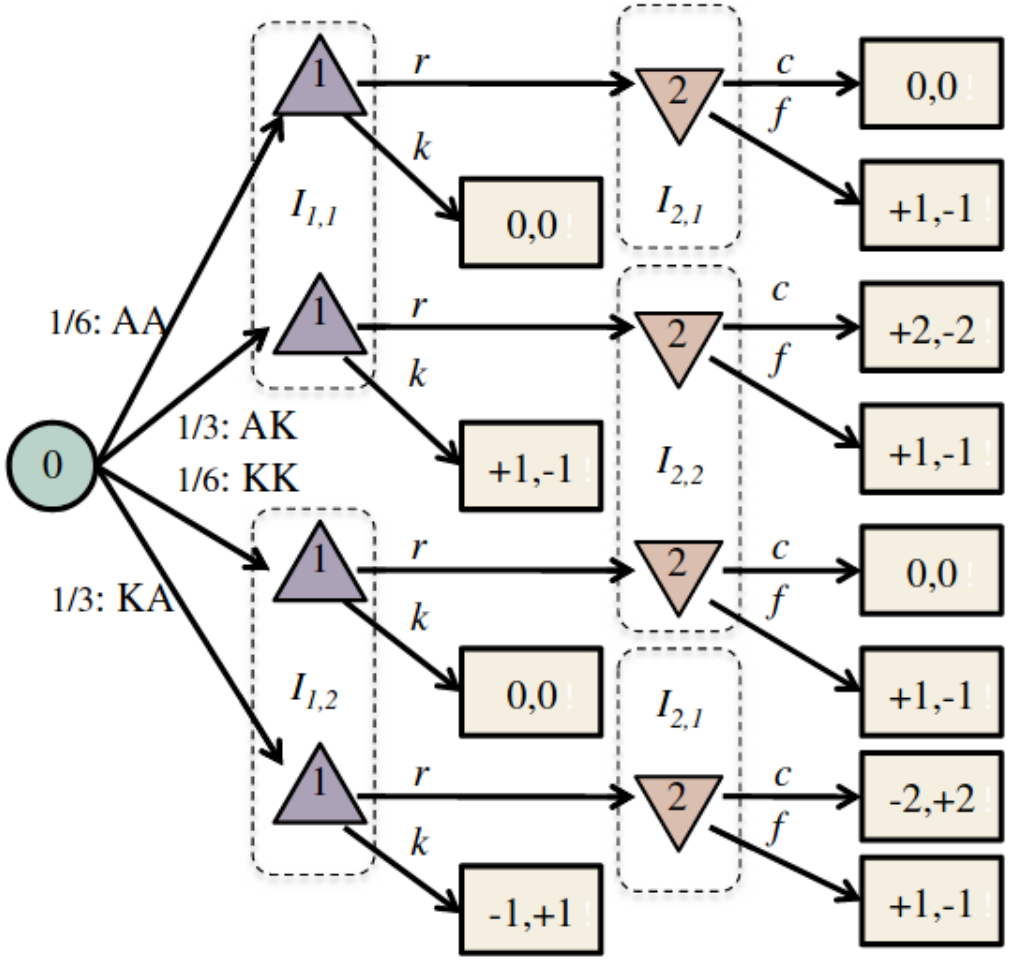
Exponential in number of info states!

Fictitious Play in Extensive Form Games

Algorithm 2 General Fictitious Self-Play

function FICTITIOUSSELFPLAY(Γ, n, m)
  Initialize completely mixed π_1
  $\beta_2 \leftarrow \pi_1$
  $j \leftarrow 2$
  while within computational budget **do**
    $\eta_j \leftarrow$ MIXINGPARAMETER(j)
    $\mathcal{D} \leftarrow$ GENERATEDATA($\pi_{j-1}, \beta_j, n, m, \eta_j$)
    for each player $i \in \mathcal{N}$ **do**
      $\mathcal{M}^i_{RL} \leftarrow$ UPDATERLMEMORY($\mathcal{M}^i_{RL}, \mathcal{D}^i$)
      $\mathcal{M}^i_{SL} \leftarrow$ UPDATESLMEMORY($\mathcal{M}^i_{SL}, \mathcal{D}^i$)
      $\beta^i_{j+1} \leftarrow$ REINFORCEMENTLEARNING($\mathcal{M}^i_{RL}$)
      $\pi^i_j \leftarrow$ SUPERVISEDLEARNING($\mathcal{M}^i_{SL}$)
    end for
    $j \leftarrow j + 1$
  end while
  return π_{j-1}
end function

function GENERATEDATA(π, β, n, m, η)
  $\sigma \leftarrow (1 - \eta)\pi + \eta\beta$
  $\mathcal{D} \leftarrow n$ episodes $\{t_k\}_{1 \le k \le n}$, sampled from strategy profile σ
  for each player $i \in \mathcal{N}$ **do**
    $\mathcal{D}^i \leftarrow m$ episodes $\{t^i_k\}_{1 \le k \le m}$, sampled from strategy profile (β^i, σ^{-i})
    $\mathcal{D}^i \leftarrow \mathcal{D}^i \cup \mathcal{D}$
  end for
  return $\{\mathcal{D}^k\}_{1 \le k \le N}$
end function



Heinrich et al. 2015 "Fictitious Self Play in Extensive-Form Games"

This slide not covered on exam